QA Algebra Practice Sheet

40 Questions | Equations, Polynomials, Surds & Indices, Logarithms, Series, Graphs, Functions, Inequalities

Name: _____ Batch: _____ Date: _____

Tiers: E = Easy, M = Moderate, X = Maximizer. Five questions per topic, in that order of difficulty. Questions with options (a) to (d) are MCQs; the rest are type-in answers.

Linear & Quadratic Equations

Q1. If α, β are the roots of $x^2 - 9x + 20 = 0$, find $\alpha^2 + \beta^2$. [E]

Q2. Find the number of non-negative integer solutions of $3x + 4y = 60$. [E]

Q3. If α, β are the roots of $x^2 - 5x + 3 = 0$, the equation with roots $1/\alpha$ and $1/\beta$ is: (a) $3x^2 - 5x + 1 = 0$ (b) $x^2 - 5x + 3 = 0$ (c) $3x^2 + 5x + 1 = 0$ (d) $x^2 - 3x + 5 = 0$ [M]

Q4. For how many integer values of k with $|k| \le 10$ does $x^2 + kx + 9 = 0$ have real roots? [M]

Q5. In how many ways can ₹100 be paid using ₹5, ₹10 and ₹20 notes, with at least one note of each kind? [X]

Polynomials & Algebraic Formulae

Q6. Find the remainder when $x^3 - 4x^2 + 5x - 7$ is divided by $(x - 2)$. [E]

Q7. If $a - b = 3$ and $ab = 10$, find $a^3 - b^3$. [E]

Q8. Find the remainder when x^{50} is divided by $(x^2 - 1)$. [M]

Q9. If $x + 1/x = 3$, find $x^5 + 1/x^5$. [M]

Q10. $p(x) = x^3 + ax^2 + bx + 6$ is divisible by $(x - 1)$ and $(x - 2)$. Find $p(3)$. [X]

Surds & Indices

Q11. Evaluate $(2^5 \times 4^3)/8^3$. [E]

Q12. If $(\sqrt{5} + \sqrt{3})/(\sqrt{5} - \sqrt{3}) = a + b\sqrt{15}$ with a, b rational, find $a + b$. [E]

Q13. If $2^x = 3^y = 6^{-z}$, find $1/x + 1/y + 1/z$. [M]

Q14. Find the value of $\sqrt{(11 + 6\sqrt{2})} + \sqrt{(11 - 6\sqrt{2})}$. [M]

Q15. How many natural numbers up to 1000 are neither perfect squares nor perfect cubes? [X]

Logarithms

Q16. Evaluate $\log_2 32 + \log_3 81 - \log_{10} 1000$. [E]

Q17. If $\log_{10} 2 = a$ and $\log_{10} 3 = b$, express $\log_{10} 72$ in terms of a and b. (a) 3a + 2b (b) 2a + 3b (c) 3a + b (d) a + 6b [E]

Q18. Find the sum of all real solutions of $\log_2(x^2 - 3x) = 2$. [M]

Q19. If $\log_4(\log_3(\log_2 x)) = 0$, find x. [M]

Q20. Find the product of all real x satisfying $x^{(\log_2 x)} = 8x^2$. [X]

Sequences & Series

Q21. Find the sum of the first 25 terms of the AP 3, 7, 11, ... [E]

Q22. Find the sum of the first 10 terms of the GP 3, 6, 12, ... [E]

Q23. The first term of an infinite GP is 12 and its sum is 16. Find the second term. [M]

Q24. Find the sum of all two-digit numbers divisible by 4. [M]

Q25. Find the value of $1\cdot2 + 2\cdot3 + 3\cdot4 + \ldots + 20\cdot21$. [X]

Graphs & Maxima-Minima

Q26. Find the minimum value of $|2x - 6| + 4$. [E]

Q27. Find the maximum value of $-x^2 + 6x - 4$. [E]

Q28. Find the area of the region $|x| + |y| \le 4$. [M]

Q29. Find the number of integer points (x, y) with $x^2 + y^2 \le 10$. [M]

Q30. Find the minimum value of $|x - 1| + |x - 4| + |x - 6| + |x - 9|$. [X]

Functions

Q31. If $f(x) = 2x + 5$, find $f(f(1))$. [E]

Q32. Find the number of integers in the domain of $f(x) = \sqrt{(x - 2)} + \sqrt{(6 - x)}$. [E]

Q33. If $f(x) = (2x + 1)/(x - 3)$, find $f^{-1}(5)$. (a) 16/3 (b) 3 (c) 5/16 (d) 14/3 [M]

Q34. If $f(x + y) = f(x) \cdot f(y)$ for all real x, y and $f(1) = 3$, find $f(4)$. [M]

Q35. f is linear with $f(0) > 0$ and $f(f(x)) = 4x + 9$. Find $f(2)$. [X]

Inequalities

Q36. Find the number of integers satisfying $|x - 3| < 5$. [E]

Q37. Find the number of integers satisfying $x^2 - 7x + 10 < 0$. [E]

Q38. Find the number of integers x in $[-5, 8]$ satisfying $(x - 2)(x + 1)/(x - 4) \geq 0$. [M]

Q39. For $x > 0$, find the minimum value of $(x + 4)(x + 9)/x$. [M]

Q40. $x, y > 0$ and $x + y = 12$. Find the maximum value of x^2y. [X]